

Course Information

Course Code: CS-11303

Credit Hours: 4 (3, 1)

Lab Hours/Week: 3

Course Title: **Operating System**

Lecture Hours/Week: 3

Pre-Requisites: **None****Design skills /techniques practiced_ Operating System_ Theory**

Students develop structured system-design thinking through:

1. Process Scheduling Design

- Designing CPU scheduling strategies
- Comparing scheduling efficiency
- Performance metric evaluation

2. Memory Allocation Modeling

- Designing paging structures
- Page replacement comparison
- Address translation design

3. Synchronization Design

- Designing semaphore-based solutions
- Critical section modeling
- Deadlock avoidance planning

4. File System Architecture Design

- Designing directory structures
- Modeling disk allocation schemes
- Evaluating disk scheduling algorithms

5. System-Level Analytical Thinking

Students learn to:

- Analyze OS trade-offs
- Evaluate performance bottlenecks
- Understand kernel vs user space separation
- Compare monolithic vs microkernel architecture